

Fig. 3.13 A projected Verlet trajectory for the pendulum model is shown in its *xy* projection for three successive intervals of timesteps (see text for discussion). The *graphs* show the transition from correct sampling behavior to motion confined to a limit cycle

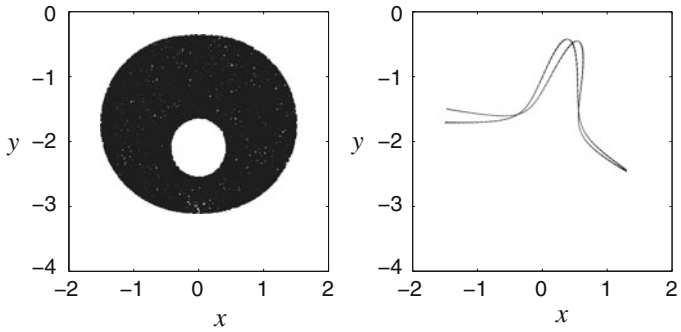


Fig. 3.14 The steady state behavior obtained in the pendulum example using Verlet (*left*) and the projected Verlet method (*right*). The graphs show the final 450,000 steps of simulation in a 500,000 step trajectory

3.6 On the Importance of Being Symplectic

Symplecticness is a strong condition and we have seen, in the construction of the perturbed Hamiltonian the last section, that it implies a constraint on the growth of errors in energy. This does not, however, automatically mean that there are no other suitable principles that could underpin the design of good numerical methods for molecular dynamics. The flow of a Hamiltonian system of the form $H = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q})$ will preserve all of the following[2]:

[2]Other conserved quantities may be present in specific situations, but this is the exception rather than the rule. For example the conservation of angular momentum, which we would expect in the case of celestial mechanics, is destroyed by the use of various boundary conditions in molecular simulation.

1. The symplectic two-form $\mathrm{d}\boldsymbol{q} \wedge \mathrm{d}\boldsymbol{p}$.
2. The Hamiltonian (i.e., the energy) H.
3. The volume in phase space (as the vector field is divergence free).
4. The time-reversal symmetry.

A natural question is which of these properties should underpin the design of a numerical method for molecular dynamics? In this section we address this through illustrative examples.

The phase volume conservation can be seen as a consequence of the symplectic property, but it is a weaker condition. It is possible to construct methods that preserve volume but which are not symplectic, and we can build methods that exactly conserve the energy (as we shall show below). Until now we have said little about the time-reversal symmetry of molecular dynamics.

3.6.1 Time-Reversal Symmetry

The term *time-reversal symmetry* or *time-reversibility* is used in two ways in connection with molecular dynamics. In this chapter, we use the term to refer to a property that can be expressed in terms of the individual trajectories of the system. (A somewhat different reversibility property is related to the evolution of probability distributions, see Chap. 8.)

Let $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$ be a dynamical system. By an *involution* we mean a linear mapping $\boldsymbol{z} \mapsto \boldsymbol{R}\boldsymbol{z}$ where $\boldsymbol{R}^2 = \boldsymbol{I}$, i.e. $\boldsymbol{R}$ is its own inverse. Given the involution $\boldsymbol{R}$ we define the *time reversal* of the vector field $\boldsymbol{f}$ with respect to $\boldsymbol{R}$ by

$$\tilde{\boldsymbol{f}}(\boldsymbol{z}) = -\boldsymbol{R}^T \boldsymbol{f}(\boldsymbol{R}\boldsymbol{z}).$$

When a vector field is its own reversal we say that it is a time-reversible vector field.

Let $H(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2 + U(\boldsymbol{q})$ be the Hamiltonian for a system of $\mathrm{N_c}$ configuration variables. Define the $2\mathrm{N_c} \times 2\mathrm{N_c}$ matrix $\boldsymbol{R}$ by

$$\boldsymbol{R} = \begin{bmatrix} \boldsymbol{I} & \boldsymbol{0} \\ \boldsymbol{0} & -\boldsymbol{I} \end{bmatrix}.$$

The vector fields involved are

$$\boldsymbol{f} = \begin{bmatrix} \boldsymbol{M}^{-1}\boldsymbol{p} \\ -\nabla U(\boldsymbol{q}) \end{bmatrix}, \quad \tilde{\boldsymbol{f}} = -\boldsymbol{R}\boldsymbol{f}(\boldsymbol{R}\boldsymbol{z}) = -\begin{bmatrix} -\boldsymbol{M}^{-1}\boldsymbol{p} \\ \nabla U(\boldsymbol{q}) \end{bmatrix},$$

which are clearly equal. Therefore the molecular dynamics Hamiltonian system is time-reversible.

The time-reversal symmetry may be understood as a property of the vector field. This engenders a symmetry at the level of the trajectories of the system.